វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា

ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ

Final Exam of Strength of Materials in 2025-2026

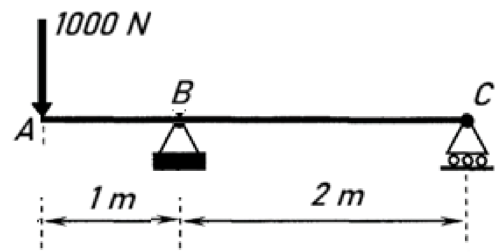
Industrial and Mechanical Egnieering Department **ECAM Engineering Program**
Date: 06 January 2026 Time: 13h00-15h00 (2 hours)

<u>Exercises [*20 pts*]</u>

1. An ABC beam is subjected to a concentrated force as shown in the figure. Properties of the beam are given as follows: Young Modulus of $E = 200\ GPa$, Poisson ratio $\nu = 0.3$, Moment of inertia of its cross-section $I = 40000\ mm^4$
   a) Sketch the deformed shape of the beam. ***[2 pts]***
   b) Determine the deflection (vertical displacement y) at mid-point of the BC-zone using this formula $EI\,y''(x) = M(x)$. ***[4 pts]***

Please respect the sign convention of internal forces and the global coordinate system from the lecture.

- $y'_{B-}(AB\ zone) = y'_{B+}\ (BC\ zone)$ (1)
- $y_B = 0$ (2)
- $y_C = 0$ (3)

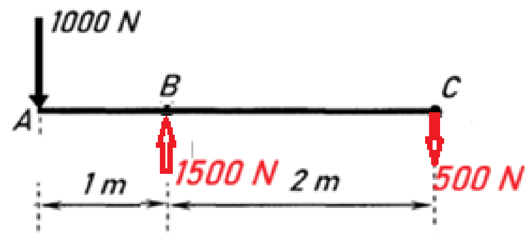


- Reaction forces:

  $R_{By} + R_{Cy} = 1000$

  $2R_{Cy} + 1000*1 = 0 \Rightarrow R_{Cy} = -500$ N

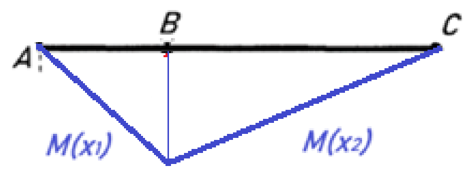
  $\Rightarrow R_{By} = 1500$ N



- Moment diagram in the two zones

  $M(x_1) = -1000*x_1$ $(0 < x_1 < 1)$

  $M(x_2) = -1000*x_2 + 1500*(x_2-1) = 500x_2 - 1500$ $(1 < x_2 < 3)$ global coordinate system



- Deflection equations in zone 2:

  $EI\, y''(x_2) = M(x_2) = 500x_2 - 1500$ $(1 < x_2 < 3)$

  $EI\, y'(x_2) = 500/2\ x_2^2 - 1500x_2 + C_1$

  $EI\, y'(x_2)= 250\ x_2^2 - 1500x_2 + C_1$

  $EI\, y\ (x_2) = 250/3\ x_2^3 - 1500/2\ x_2^2 + C_1x_2 + C_2$

  $EI\, y\ (x_2) = 250/3\ x_2^3 - 750\ x_2^2 + C_1x_2 + C_2$

  - $y_B = 0$ (2) $\Leftrightarrow$ $EI\, y_B = 0$ $\Leftrightarrow$ $EI\, y(x_2{=}1) = 0$

    $250/3\ (1)^3 - 750\ (1)^2 + C_1(1) + C_2 = 0$

    $C_1 + C_2 = 750 - 250/3 = 2000/3$ (4)

  - $y_C = 0$ (3) $\Leftrightarrow$ $EI\, y_C = 0$ $\Leftrightarrow$ $EI\, y(x_2{=}3) = 0$

    $250/3\ (3)^3 - 750\ (3)^2 + C_1(3) + C_2 = 0$

    $3C_1 + C_2 = 750\ (3)^2 - 250*(3)^2 = 500*9 = 4500$ (5)

    (5)-(4) => $2C_1 = 4500 - 2000/3 = 11500/3$ => $C_1 = 5750/3$

    (4)=> $C_2 = 2000/3 - 5750/3 = -3750/3$

  Finally,

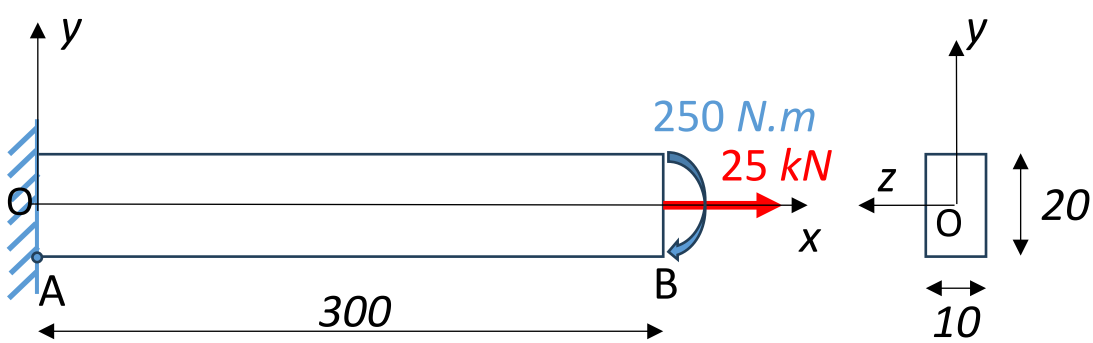
  $EI\, y\ (x_2) = 250/3\ x_2^3 - 750\ x_2^2 + 5750/3 x_2 - 3750/3$

- Deflection at mid-BC is $y(x_2{=}2)$

EI y(x_2=2) = 250/3 $(2)^3$–750 $(2)^2$ + 5750/3(2) -3750/3=250

=> y(x_2=2) = 250/(EI) = 250/(200e9*40000e-12) = 3.12e-2 *m* or **31.2 *mm***

2. A cantilever beam is subjected to combined loadings of tension (*25 kN*) and bending (*250 N.m*). Please use **superposition** between tension and bending for the following questions.
   a) Determine the normal stresses (σ_x and σ_y) at point *A* (bottom left of the beam) ***[4 pts]***
   b) Determine the shear stress (τ_{xy}) at point A ***[2 pts]***
   c) Please draw state of stress at point A ***[2 pts]***
   d) Please draw a Mohr circle representing the state of stress at point A. ***[2 pts]***
   e) Determine the maximum normal stress in tension (σ_x^{Max}) ***[2 pts]***
   f) Specify the locations (*x =? mm, y=? mm, z = ? mm*) of the maximum normal stress found in question c) as reference to the chosen global coordinate system. ***[2 pts]***



a) normal stresses (σ_x and σ_y) at point *A*

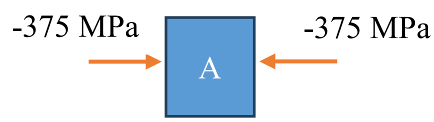
- In tension: σ_x = F/A = 25000/(10*20) = 125 MPa
- In bending: σ_x= -M(x=0)*y_A/I_{oz} = -(-250*10e3)*(-10)/(10*20^3/12) = -250*12/8
  M(x) = -250 N.m (constant diagram along x-axis and negative by sign convention)
  σ_x = -375 MPa
- Total at A: σ_x = 125-375 = -250 MPa; σ_y = 0 MPa

b) shear stress τ_{xy} at point *A*

- In tension: τ_{xy} = 0 MPa
- In bending: τ_{xy}= V_y(x)*Q_{oz}(y=-10)/(I_{oz}*b) = 0 MPa
  Because V_y(x) = 0 and Q_{oz}(y=-10)=0

- In total at A: τ_{xy}= 0 MPa

c) State of stress at A



d) Mohr Circle of state of stress at point A

e) Maximum normal stress in tension: σ_x^{Max}=125 MPa + 375 MPa = 500 MPa

f) Location

- 0< x <300 mm; y = 10 mm; -5 mm < z <+5 mm: everywhere along the top surface of the beam.